

Jacky Kam

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WORK EXPERIENCE

Mechanical Engineering Intern, Lab Technician - Xtract One Technologies

July 2022 – June 2024 | Vancouver, BC

- Designed an automated data collection system in Onshape, achieving data consistency of $\pm 0.5\%$ across 20,000 samples using stepper motors coded in Python and Q Programmer.
- Improved the detection accuracy of small threat objects by 60% through prototyping a proof-of-concept hybrid passive / active detection system using custom 3D-printed mountings to integrate existing components.
- Optimized sensor response by 500% through the research and implementation of magnetic bucking concepts to divert magnetic fields from sensor coils.
- Streamlined assembly and setup times in the field by 30% through procurement of technical documentation on the BOM, assembly, maintenance, and diagnosis of electronics and software.
- Patent Pending: System and Method of Object Classification and Detection Using a Magnetic Property Array-Based Object Detection System. PCT/CA2023/050459.

ENGINEERING DESIGN TEAMS

Payload Lead - UBC Uncrewed Aircraft Systems

September 2022 – June 2023 | Vancouver, BC

- Coached a team of 20+ interdisciplinary students to cultivate creative design, CAD and manufacturing competencies in the context of designing a scaled-down realistic air transportation system.
- Guided students through a carbon-fibre layup process by researching matrix composites, manufacturing techniques, performing CFD analysis, and validating findings using a wind-tunnel.
- Led the design of a non-sequential winch system using electromagnetic clutches to meet impact constraints of $< 2\text{m/s}$ in a 30ft radius when deployed from 75+ ft AGL.

Payload Designer, Integration Specialist - UBC Uncrewed Aircraft Systems

November 2021 – June 2022 | Vancouver, BC

- Independently designed a lightweight package retrieval system in SolidWorks utilizing linear actuators to carry 2kg payloads, increasing package retrieval consistency by 80%.
- Optimized drone maneuverability over 4 iterations through the FEA, prototyping and manufacturing of waterjet-cut aluminum landing gear, allowing mecanum drive, reducing weight by 40%, and assembly time by 50%.

TECHNICAL PROJECTS

Mechanical Designer, Integration Specialist - Haptic Manipulator Glove

September 2023 – April 2024 | Vancouver, BC

- Developed an under-actuated finger linkage assembly for use on a custom End-of-Arm Tool in SolidWorks, using linear actuators to emulate finger motion with 150-degree range of movement and 2mm accuracy.
- Iteratively optimized the design of waterjet-cut aluminum finger linkages over 5+ revisions to minimize material usage and reduce linkage play, achieving tolerances of $\pm 0.1\text{mm}$.
- Designed and integrated a hand + forearm-mounted controller glove to track curling of user's fingers with $< 2\text{ms}$ latency, using surface modelling to balance aesthetics and reduce weight.

Software Lead, Integration Specialist – Tremor Stabilization Glove

September 2021 – April 2022 | Vancouver, BC

- Leveraged Python (SKLearn) to discern user tremor from EMG readings and modulate flywheel to mitigate tremors from the hand, resulting in a detection accuracy of 95%, and startup response time of < 0.5 seconds.
- Designed and manufactured a vibration dampening enclosure in SolidWorks over 3 iterations. Accommodated for precession of flywheel and motor, while considering a compact footprint, ease of maintenance, and safety.

EDUCATION

University of British Columbia - BAsC Integrated Engineering

September 2019 – May 2024 | Specializing in Mechanical and Materials Engineering

SKILLS

Manufacturing – Additive Manufacturing (FDM/SLA), Rapid Prototyping, MIG/TIG/Gas Welding, Waterjet, Soldering

Software – SolidWorks, Onshape, MATLAB, Arduino, RStudio, GitHub, Microsoft Office

Languages – C, Python, JavaScript, HTML/CSS, Assembly, VHDL